

SparkBridge Provider

Provider is the broker control and policy layer: an embedded Moquette MQTT broker for zero-config single-node use, or a companion configuration for an external HiveMQ CE broker with the SparkBridge authorization extension.

Version 3.0.0Ignition 8.1.19+ and 8.3Runs on: the gateway hosting the broker
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Overview

1. Overview

Provider is the policy and routing layer for SparkBridge's Sparkplug traffic. It carries plant data between sites and consumers while enforcing identity, permission and connection boundaries; Sparkplug itself leaves security to the implementation, and the broker is where that posture actually gets set. Provider ships two broker shapes: an embedded Moquette broker on loopback for a single node with nothing to install, or a companion configuration for an external HiveMQ CE broker with the SparkBridge broker-authorization extension for MQTT 5, shared subscriptions and Sparkplug Aware certificates. Neither shape is licensed by connected-client count.

Provider is not itself a tag provider or a decoder; Host does that. Install Provider first, before Host and Edge.

Requirements

2. What You Need

Ignition	8.1.19 or newer on the 8.1 line, or 8.3 (separate build). A gateway that can be restarted at will.
Java	Embedded broker: none beyond Ignition's own runtime. External HiveMQ CE: a JDK 11 or newer on the broker host.
Gateway role	Any gateway that will host the embedded broker, or the gateway running alongside an external HiveMQ CE instance.

Network	Embedded broker binds 127.0.0.1:1883 plaintext and, with TLS enabled, 127.0.0.1:8883, with no sparkbridge.* system properties set.
Other SparkBridge modules required	None for the embedded broker. For external HiveMQ CE: the SparkBridge broker authorization extension and the HiveMQ file-RBAC extension on the broker host.

Install

3. Install

1. Download the release bundle for your Ignition line from the [download page](#): SparkBridge-Provider.mod1, SparkBridge-Host.mod1, SparkBridge-Edge.mod1, plus SHA256SUMS.txt and sparkbridge-signing-cert.pem.
2. Install order matters: Provider first, before Host and Edge. In the gateway, go to Config > Modules > Install or Upgrade a Module and upload SparkBridge-Provider.mod1 first.
3. Accept the certificate and license once; confirm Config > Modules shows SparkBridge Provider state = Running.

With no sparkbridge.* system properties set, the Provider binds the embedded Moquette broker on 127.0.0.1:1883 and, with TLS enabled, 127.0.0.1:8883. If pairing with an external HiveMQ CE broker, start HiveMQ before the Ignition gateway every time: both want ports 1883 and 8883, and whichever starts first takes the ports.

Upgrade every installed SparkBridge module together, from the same release bundle, in the same maintenance window, to avoid mixed-version disagreement over shared settings.

License

4. License

File	Per-gateway license key at <data>/sparkbridge-license.key.
Re-check interval	Provider re-checks the key file every 30 seconds. Installing, replacing or removing a key takes effect with no restart.

From release 2.7.0, Provider runs on Ignition's standard two-hour module trial (the gateway's Reset Trial button restarts the clock). When the trial expires with no verified key naming the product, the embedded broker refuses new connections while established sessions are left alone. A key dropped in after expiry lifts the pause on its own within the 30-second re-check, no button press and no restart. A verified key overrides the trial entirely.

Configure

5. Configure

Primary path: open Config > SparkBridge Provider > Settings. Two keys on this page are editable, both Provider-owned and both read once at startup (restart to apply a change):

broker.port	Embedded Moquette plaintext listener port. Default 1883.
broker.tlsport	Embedded Moquette TLS listener port. Default 8883.

Everything else on the page is a read-only audit of the broker's effective running posture: implementation, broker endpoints, bind host, TLS, plaintext, client authentication, keystore, identities, edge scopes, application topics, and secrets status. Secrets are never shown in the clear; a configured broker password renders only as a boolean, for example `Broker password configured: true`.

Broker-extension RBAC (external HiveMQ CE only)

Configure the separate broker-extension on the broker host. Its file-based RBAC uses `extensions/sparkbridge-auth/sparkbridge-auth.properties` for `sparkbridge.broker.hostuser`, `sparkbridge.broker.edgescopes`, `sparkbridge.broker.hosttopics`, `sparkbridge.broker.apptopics`, and `sparkbridge.broker.appprefixes`. HiveMQ's `credentials.xml` carries the broker credentials and password hashes. Keep its identities and scopes aligned with the Provider settings page.

Operate

6. Operate

Provider needs no day-to-day attention once running; watch Config > SparkBridge Provider > Settings under Posture for the effective running values, since a property that does not appear there indicates a configuration gap.

Security posture: an edge credential is bound to one group and node, an unlisted edge is denied, and edge identities cannot publish NCMD, DCMD or STATE at all (structurally unavailable, proven against the deployed ACL). On the SparkBridge broker, UNS, file-transfer and bridge application topics are authorized per identity by prefix and denied by default. TLS hostname verification is mandatory; disabling it requires an explicit development-only flag with no production use.

On restart, an embedded broker rebinds its configured ports; established client sessions reconnect on their own retry logic. On link loss to an external HiveMQ broker, Provider's own audit page reflects the broker as unreachable until it reconnects.

Verify

7. Verify

Embedded broker, gateway log:

```
SparkBridge Provider: ... broker
```

External broker, gateway log:

```
SparkBridge Provider: broker.impl='external', embedded broker disabled
```

Config > SparkBridge Provider > Settings, under Posture, shows the running values rather than the file's intent.

Troubleshoot

8. Troubleshoot

Symptom	Likely cause	Fix
External HiveMQ fails to start	Ignition gateway started first and took ports 1883/8883	Start HiveMQ before the Ignition gateway, every time
Edge cannot connect	Edge credential not bound to that group/node, or unlisted	Add the edge identity to the Provider's identity/scope configuration
Posture page missing an expected property	Configuration gap	The absence itself is the signal; add the missing setting
New connections refused with existing sessions unaffected	Module trial expired, no verified key	Drop a valid <code>sparkbridge-license.key</code> in the data directory

Symptom	Likely cause	Fix
Broker password shows only as a boolean	Expected behavior, not a bug	Confirm the value in the secrets file directly if verification is needed
Reference		

9. Reference

License file	<data>/sparkbridge-license.key
Module file	SparkBridge-Provider.modl
Embedded broker default ports	1883 plaintext, 8883 TLS
Broker-extension config file	extensions/sparkbridge-auth/sparkbridge-auth.properties
License re-check	30 seconds
Support	

10. Support

Full install and config detail: [SparkBridge Docs, Provider](#). Product overview: [Provider on the products page](#).
 Questions: [Contact](#). Release history: [GitHub releases](#).